

ADAPTIVE CIVIC ROUTING INTELLIGENCE SYSTEM

Report once. Route correctly. Resolve with accountability.

CORE OPERATIONAL VALUE PROPOSITION

"Location determines responsibility. AI assists. GIS decides."

ACRIS eliminates municipal boundary ambiguity and manual triage latency by transforming citizen coordinates into deterministic municipal department assignments via PostGIS spatial containment, while Google Gemini assists solely with unstructured complaint categorization.

THE MUNICIPAL ROUTING PROBLEM IN INDIAN URBAN GOVERNANCE

Civic complaints suffer multi-week delays not from field lethargy, but from **jurisdiction ambiguity**: street names, postal PINs, and colloquial landmarks do not map cleanly to administrative multi-polygons.

TRADITIONAL FAILURE

Manual triage backlogs (48-72h), subjective guesswork, ticket bouncing across departments, and unexplainable routing history upon ward realignments.

ACRIS INNOVATION

Mathematical PostGIS containment, versioned spatial cadastre, instant Socket.IO SLA dispatch, and permanent append-only audit ledgers.

CIVIC INTELLIGENCE SEPARATION OF CONCERNS

GEMINI ASSISTS (AI)

Parses unstructured Kannada/English text, extracts issue tags, marks **AI_SUGGESTED** category. Zero database permissions.

POSTGIS DECIDES (GIS)

Executes **ST_Covers()** on active MultiPolygons for deterministic GIS jurisdiction resolution municipal queue dispatch.

MYSURU CIVIC CADASTRE • EPSG:4326 MUNICIPAL JURISDICTIONS



Mysuru City Corporation (MCC) & MUDA Jurisdictional Engine

Municipal Jurisdiction Boundaries • Deterministic Routing • Deterministic GIS Resolution

ACTIVE CADASTRE: MYS_2026_V2

POSTGRESQL + POSTGIS

FULLY AUDITABLE

DETERMINISTIC SPATIAL ENGINE

PostgreSQL + PostGIS

GiST R-Tree spatial indexing on administrative MultiPolygons delivers efficient point containment queries.

SPATIAL PREDICATE STANDARD

ST_Covers(j.boundary, pt)

Strict binary inclusion prevents border ambiguity; handles exact edge incidents with jurisdiction consistency.

ACRIS END-TO-END CIVIC INTELLIGENCE PIPELINE

✓ Zero Manual Queueing • Full Cryptographic Traceability

01 CITIZEN

Intake Report

PWA / Mobile Capture

Photo + Pin

02 LOCATION

GPS Ingestion

Lat/Lng Coordinates

EPSG:4326 Point

03 POSTGIS

Spatial Query

Binary Containment

ST_Covers()

04 AUTHORITY

Jurisdiction

Jurisdiction Entity

MCC vs MUDA

05 DISPATCH

Dept Queue

Automated Binding

MCC_ROADS

06 OPERATOR

Realtime Sync

Socket.IO Broadcast

48h SLA Clock

07 RESOLUTION

Field Closure

Post-Repair Proof

Photo Proof

08 AUDIT

Immutable Log

Sealed Ledger Row

SHA-Stamped

Why Civic Complaints Get Stuck: The Spatial Ambiguity Breakdown

"A citizen can report an issue correctly — yet the complaint can still reach the wrong authority."

THE REAL-WORLD FAILURE PIPELINE: HOW CIVIC COMPLAINTS GET MISROUTED

Status Quo: Up to 35% of boundary-adjacent tickets delayed or misdirected



01 / SPATIAL JURISDICTION

Boundary Ambiguity

Citizens describe incidents using street names or landmarks ("near Sayyaji Rao Road"). Arterial corridors cross multiple wards and agencies with overlapping jurisdictions.

```
-- Street name SQL search fails:
SELECT * FROM wards
WHERE street_name LIKE '%Sayyaji%'
-- Error: Matched 3 distinct wards!
```

FAILURE IMPACT

Text-based search cannot resolve which side of a road belongs to which authority. 35% of boundary tickets are initially misrouted.

Root Cause: Administrative boundaries are irregular multi-polygons, while street text lacks spatial topology.

ACRIS SOLUTION

Deterministic MultiPolygon Match

PostGIS ST_Covers(j.boundary, pt) tests exact coordinate against spatial polygons, eliminating border ambiguity completely.

02 / ADMINISTRATIVE BOTTLENECK

Manual Routing Triage

Complaints enter a centralized triage queue where human operators must read text descriptions, inspect static boundary PDFs, and manually select a maintenance department.

```
// Manual triage backlog state:
ticket.status = 'PENDING_TRIAGE';
ticket.assigned_to = null;
// Queue delay: 48 to 72 hours!
```

FAILURE IMPACT

During monsoon rains, helplines are overwhelmed. Operators make hurried subjective guesses, causing extensive triage backlogs.

Root Cause: Lack of automated spatial routing forces human operators to become manual routing intermediaries.

ACRIS SOLUTION

Instant Spatial Index Dispatch

GIST R-Tree indexing evaluates containment in <15ms. Ingestion immediately maps coordinates to department queues (MCC_ROADS).

03 / INTER-AGENCY FRICTION

Wrong Dispatch & Bouncing

When an unverified complaint reaches MCC instead of MUDA, field engineers reject it upon arrival. The ticket bounces back to triage, resetting SLA clocks while the defect remains open.

```
// Inter-agency ping-pong:
MCC_ROADS -> rejects ("MUDA Road")
MUDA_MAINT -> rejects ("Core Ward")
// Result: SLA reset, defect remains!
```

FAILURE IMPACT

Departments deny maintenance liability across municipal boundary peripheries. Citizen receives multiple contradictory SMS updates.

Root Cause: No single spatial source of truth exists between urban local bodies (MCC) and urban development authorities (MUDA).

ACRIS SOLUTION

Jurisdiction Boundary Lock

Department ownership is mathematically proven prior to work order creation. Socket.IO pushes tickets directly to verified field dashboards.

04 / TEMPORAL GOVERNANCE

Lost Historical Provenance

Municipal wards are periodically redrawn, realigned, or renamed. In conventional databases, updating boundary tables overwrites past geometry, corrupting legal provenance.

```
-- Naive database update:
UPDATE boundaries SET geom = $new;
-- Destroys legal history of past tickets!
-- Historical audits fail in tribunal
```

FAILURE IMPACT

Municipal audits, court litigation, and civic compliance inquiries fail for past fiscal years because past routing context cannot be reconstructed.

Root Cause: Absence of versioned cadastre. Boundary mutations overwrite historic spatial records.

ACRIS SOLUTION

Temporal Cadastre Versioning

Complaints permanently store jurisdiction_version_id. Historical records link immutably to MYS_2026_V1, while new tickets resolve against active MYS_2026_V2.

THE CENTRAL ENGINEERING QUESTION SOLVED BY ACRIS

"How do we determine the responsible authority from location — reliably and reproducibly?"

DETERMINISTIC POSTGIS SPATIAL CONTAINMENT

The Same Issue. Different Authorities. Location Determines Responsibility.

A pothole on Sayyaji Rao Road vs Outer Ring Road requires completely different municipal dispatch.

Point in Sayyaji Rao Rd (Urban Core)

Central Mysuru Commercial Corridor

MCC JURISDICTION



CANONICAL INCIDENT: HM-CIV-2026-000656

COORDINATE FIX:

Sayyaji Rao Rd / MCC Central Zone

ACTIVE CADASTRE:

MYS_2026_V2 (ACTIVE)

MATCHED ZONE:

MCC_ZONE_1 (MCC Central Zone)

GEMINI ROLE:

POTHOLE (AI_SUGGESTED)

DETERMINISTIC ROUTING RESULT

MCC Roads & Infrastructure (MCC_ROADS)

Zero Human Triage

Point in Northwest Sector (MUDA Layout)

Outer Ring Road Peripheral Corridor

MUDA JURISDICTION



PERIPHERAL ROAD CORRIDOR: RING ROAD NORTHWEST

COORDINATE FIX:

Outer Ring Rd / Northwest Sector

ACTIVE CADASTRE:

MYS_2026_V2 (ACTIVE)

MATCHED ZONE:

MUDA_SECTOR_NW (MUDA Layout)

GEMINI ROLE:

POTHOLE (AI_SUGGESTED)

DETERMINISTIC ROUTING RESULT

MUDA Maintenance Division (MUDA_MAINT)

Zero Human Triage

END-TO-END DETERMINISTIC TRANSFORMATION PIPELINE

01 INTAKE

Citizen Input

Photo + Coordinates + Text

(lng, lat, description)

02 AI ASSIST

Gemini Suggestion

Tag Extraction only

tag: POTHOLE (non-binding)

03 SPATIAL GIS

PostGIS Predicate

Binary Containment

ST_Covers(boundary, pt)

04 RESOLUTION

Jurisdiction Lock

Municipal Cadastre

MCC_ZONE_1 (V2 ACTIVE)

05 DISPATCH

Dept Work Queue

Automated Binding

MCC_ROADS (48h SLA)

06 AUDIT

Sealed Record

Permanent Trace

INSERT INTO audit_logs

✓ Zero Text Guessing • Coordinates Map Deterministically to Municipal Authority

CORE PRINCIPLE: "LOCATION DETERMINES RESPONSIBILITY."

Both reports have identical complaint descriptions — but spatial containment routes them to completely different legal authorities.

From Citizen Report to Audited Closure: Complete 6-Stage Execution

A seamless 6-stage pipeline delivering automated spatial routing, operator SLA tracking, and sealed audit closure.

STAGE 01

PWA

Citizen Intake

Citizen opens mobile PWA, snaps photographic proof, positions interactive Leaflet pin, and provides informal description.

- Camera photo upload via Multer buffer
- Leaflet OpenStreetMap draggable pin
- Duplicate detection (<50m radius)

```
navigator.geolocation
Leaflet.marker(lng, lat)
```

STAGE 02

EXPRESS

Validation & AI

Node.js backend validates coordinate bounds, verifies user authentication, and invokes Gemini for category extraction.

- Express payload & schema check
- JWT authentication in secure cookie
- Gemini suggests: AI_SUGGESTED tag

```
POST /api/complaints
tag: AI_SUGGESTED (pothole)
```

STAGE 03

POSTGIS

GIS Resolution

PostGIS constructs standard geometry point and evaluates binary containment against active versioned MultiPolygons.

- ST_SetSRID(ST_MakePoint, 4326)
- GiST R-Tree index bounding search
- Evaluates WHERE is_active = true

```
ST_Covers(j.boundary,
ST_SetSRID(pt, 4326))
```

STAGE 04

ROUTING

Jurisdiction Match

Coordinates bind to municipal jurisdiction entity. System automatically maps to corresponding department queue.

- Resolves: MCC Central Zone
- Active Cadastre: MYS_2026_V2
- Department: MCC_ROADS

```
JOIN jurisdictions j
ON department_id = MCC_ROADS
```

STAGE 05

REALTIME

Operator & Sync

Socket.IO broadcasts new ticket to operator dashboard in real-time. Active 48-hour SLA countdown timer begins immediately.

- WebSockets push to dept room
- Live 48h SLA countdown active
- Operator claims work order

```
io.to('dept:MCC_ROADS')
.emit('ticket:created')
```

STAGE 06

AUDIT

Resolution & Audit

Field crew uploads post-repair photograph. Status updates to RESOLVED. Append-only audit record permanently sealed.

- Resolution photo proof required
- Citizen notified via WebSocket
- Locked to version MYS_2026_V2

```
INSERT INTO audit_logs
action: 'RESOLVED' (sealed)
```

DATA INTEGRITY & FAIL-SAFE GUARANTEES

Robust Ingestion Architecture

Leaflet Map Pinning:

If GPS hardware fix is noisy (>15m), citizen can manually drag the pin marker on OpenStreetMap tiles to exact defect.

Gemini Fail-Open:

If Google Gemini API is throttled or offline, complaint ingestion proceeds unimpeded using citizen-selected fallback category.

Boundary Screening:

Points falling outside all active MultiPolygons route to an unassigned queue with automatic operator alerts.

```
marker.on('dragend', (e) => updateLocationCoords(e.target.getLatLng()));
```

✔ Zero complaint loss under network, API, or hardware degradation.

REAL-TIME SLA TRACKING & DISPATCH

Real-Time Municipal SLA Tracking

Indexed Spatial GIS Query:

PostGIS ST_Covers evaluates containment in <15ms via PostgreSQL GiST spatial R-Tree indexing.

Socket.IO Room Broadcasting:

Complaints instantly emit to department-specific WebSocket rooms (dept:MCC_ROADS), eliminating 72-hour triage backlogs.

SLA Clock Commitment:

Active 48-hour emergency asphalt patch SLA countdown begins immediately upon database transaction commit.

```
io.to('dept:MCC_ROADS').emit('ticket:created', { id, sla: '48h' });
```

✔ Automated queue placement cuts initial triage from 72 hours to <1 second.

HUMAN OVERSIGHT & AUDIT SEAL

Accountable Closure with Human Review

Human Reassignment Oversight:

Authorized operators can override routing only by entering mandatory justification notes stored in the audit log.

Immutable Version Stamping:

Every routing decision permanently locks the foreign key of the active cadastre version (MYS_2026_V2).

Field Photo Proof:

Closure requires after-repair photographic proof uploaded by municipal field teams, notifying citizen via WebSockets.

```
INSERT INTO audit_logs (complaint_id, action, notes) VALUES ($1, 'RESOLVED', ...);
```

✔ fully auditable provenance for accountable municipal workflows.

HUMAN IN THE LOOP

Human oversight remains active after deterministic GIS routing. Operators can reassign with mandatory audit justification notes.

✔ Zero Unsupervised LLM Execution

AI Assists. GIS Decides.

Strict architectural separation between stochastic natural language understanding and deterministic municipal jurisdiction.

GOOGLE GEMINI API

ASSISTIVE INTELLIGENCE ONLY

Natural Language Understanding

Interprets informal citizen descriptions, extracts structured issue properties, and accelerates reporting intake:

- **Category Suggestion:** Recommends tags (POTHOLE, DRAINAGE, STREETLIGHT).
- **Attribute Extraction:** Parses severity, hazard hints, and contextual entities.
- **Informational Flag:** Categorization is explicitly tagged as **AI_SUGGESTED**.
- **Friction Removal:** Citizens avoid confusing municipal departmental drop-downs.

STRICT SAFETY BOUNDARY:

Gemini has ZERO access to routing tables or boundaries. Gemini NEVER decides municipal jurisdiction.

```
{
  "suggested_category": "POTHOLE",
  "confidence_score": 0.94,
  "classification_tag": "AI_SUGGESTED",
  "jurisdiction_authority": null
}
```

GEMINI OPERATIONAL BOUNDARIES

✓ PERMITTED:

Issue tag extraction
Text summarization
Severity estimation

✗ FORBIDDEN:

Selecting department
Determining ward
Modifying database

Fail-Open Fallback: If Gemini is unreachable, intake falls back seamlessly to citizen manual tag.

Execution Model: Asynchronous NLP classification; zero database read/write permissions.

Status: Non-binding suggestion. Operator can override at any time.

VS

POSTGRESQL + POSTGIS

DETERMINISTIC SPATIAL ENGINE

Municipal Jurisdictional Authority

Municipal responsibility is a legal question governed by territorial geometry. PostGIS provides deterministic GIS jurisdiction resolution:

- **Point Construction:** `ST_SetSRID(ST_MakePoint(lng, lat), 4326)`.
- **Containment Evaluation:** `ST_Covers(j.boundary, point)`.
- **Cadastre Filtering:** Only queries active cadastre versions (`is_active = true`).
- **Department Mapping:** Direct join from spatial polygon to responsible department queue.

MUNICIPAL CADASTRE AUTHORITY:

Jurisdiction is governed by gazetted boundaries. PostGIS delivers reproducible polygon math.

```
SELECT j.id, j.name, j.department_id FROM jurisdictions j
JOIN jurisdiction_versions jv ON j.version_id = jv.id
WHERE jv.is_active = true
  AND ST_Covers(j.boundary, ST_SetSRID(ST_MakePoint($1, $2), 4326));
```

POSTGIS SPATIAL GUARANTEES

Topological Predicate:

`ST_Covers` strict boundary closure; exact vertex edge inclusion.

GiST Indexing:

Efficient R-Tree candidate pruning; audit-ready defensibility.

Zero Stochastic Drift: Mathematical geometry guarantees identical output for identical coordinates.

Execution Model: Deterministic spatial lookup; audit log foreign key committed.

Status: Legally binding spatial resolution committed to database audit log.

"Generative AI interprets the complaint. Spatial GIS determines responsibility."

Core Principle: AI is assistive. Municipal jurisdiction is strictly deterministic PostGIS math.

How the System Works Underneath: Full-Stack Architecture

A production-ready full-stack architecture with spatial intelligence, real-time sync, and cryptographic auditability.

TIER 1 / CLIENT LAYER **React 18 + Vite SPA & PWA Engine**

DEPLOYED ON VERCEL EDGE CDN

Citizen Portal (Mobile PWA)

Leaflet 1.9 map picker, camera capture, duplicate detection <50m, mobile-first responsive UI.

HTML5 Geolocation + OpenStreetMap

Operator Workspace

Real-time Socket.IO incoming queue, 48h SLA countdown timers, department filtering, human review.

Socket.IO Client + 48h SLA Timer

Admin Cadastre Console

Leaflet GeoJSON boundary viewer, cadastre version management (MYS_2026_V1/V2), audit trail.

GeoJSON Visualizer + Audit Ledger UI

▼ REST API HTTP(S) (Multipart Form / JSON) + WebSocket Bidirectional Sync ▼

TIER 2 / API GATEWAY **Node.js 20 + Express Routing Engine**

DEPLOYED ON RENDER

Routing Service (routingService.js)

Constructs spatial points, calls PostGIS containment, handles active version resolution.

Security & Middleware

JWT auth in cookies, RBAC role gating (CITIZEN, OPERATOR, ADMIN), Multer photo uploads.

Endpoints: POST /api/complaints • GET /api/jurisdictions • POST /api/auth • GET /api/audit

EVENT STREAM ENGINE

Socket.IO Realtime Sync

WebSocket engine with automatic reconnection. Subscribes operator clients to department rooms for zero-latency ticket push.

```
io.to('dept:MCC_ROADS')
  .emit('ticket:created', ticket);
```

REAL-TIME DISPATCH

▼ PostgreSQL pg.Pool Connection (Parameterized SQL Queries & GiST Optimization) ▼

TIER 3 / PERSISTENCE & SPATIAL STORE **PostgreSQL + PostGIS Spatial Database**

MANAGED ON RENDER CLUSTER

Spatial Cadastre Tables

MultiPolygon boundaries stored in EPSG:4326. GiST R-Tree spatial indexing delivers sub-ms point checks.

Table: jurisdictions (MultiPolygon)

Temporal Versioning Engine

jurisdiction_versions maintains active cadastre flags (MYS_2026_V1 RETIRED vs MYS_2026_V2 ACTIVE).

Table: jurisdiction_versions

Immutable Audit Ledger

audit_logs append-only table recording initial dispatch, operator reassignment justifications, and photo proof.

Table: audit_logs (append-only)

FRONTEND DEPLOYMENT

Vercel Edge Global CDN

SSL / HTTP2

BACKEND RUNTIME

Render Linux Managed Service

NODE 20 LTS

SPATIAL DATABASE

Render PostgreSQL + PostGIS

GIST R-TREE

DATA FLOW: Citizen GPS/Photo → Express Auth & Validation → Gemini Categorization (Assistive) → PostGIS ST_Covers (Deterministic) → Socket.IO Operator Queue → Permanent Audit Record

PRODUCTION READY & DEPLOYED

Municipal Boundaries Change. History Must Not.

When municipal wards are realigned or renamed, past routing decisions must remain explainable and auditable.

MUNICIPAL BOUNDARY LIFECYCLE: THE REDISTRICTING TIMELINE

CADASTRE VERSION: MYS_2026_V1

RETIRED

GAZETTE NOTIFICATION



Ward Realignment

CADASTRE VERSION: MYS_2026_V2

ACTIVE

Valid: 2026-04-01 to PRESENT • Realigned Ward Demarcations
All new incoming complaints evaluated against this active geometry.

Valid: 2026-01-01 to 2026-03-31 • Original Municipal Demarcation
Historic complaints permanently linked to this version geometry.

HISTORICAL COMPLAINT (DEMARCATION V1)

FILED FEB 2026

Historical Complaint (Demarcation V1)

Filed prior to April 2026 redistricting. Must remain permanently explainable under the exact boundary rules active when reported.

- **Routing Timestamp:** February 12, 2026 (Historic Ingestion).
- **Linked Version:** jurisdiction_version_id = MYS_2026_V1.
- **Resolved Authority:** MCC Central Zone (under V1 demarcation).
- **Audit Integrity:** Even if boundaries shift today, re-running historical audit reproduces the exact same legal authority that had jurisdiction in February.

```
SELECT c.id, c.created_at, j.name AS assigned_zone, jv.version_code
FROM complaints c
JOIN jurisdictions j ON c.jurisdiction_id = j.id
JOIN jurisdiction_versions jv ON c.jurisdiction_version_id = jv.id
WHERE jv.version_code = 'MYS_2026_V1';
```

✓ Historical dispatch remains 100% explainable in court or civic audit.

NEW COMPLAINT (ACTIVE CADASTRE V2)

FILED UNDER ACTIVE CADASTRE

Complaint: HM-CIV-2026-000656 (Sayyaji Rao Rd)

Filed after April 2026 redistricting. Evaluates strictly against active boundary multi-polygons.

- **Routing Timestamp:** Current Operational Date (Live Ingestion).
- **Linked Version:** jurisdiction_version_id = MYS_2026_V2.
- **Resolved Authority:** MCC Central Zone (MCC_ROADS queue).
- **Dynamic Containment:** Evaluated strictly against polygons where is_active = true. Historical boundaries are ignored for active routing.

```
SELECT j.id, j.name, j.department_id
FROM jurisdictions j
JOIN jurisdiction_versions jv ON j.version_id = jv.id
WHERE jv.is_active = true
AND ST_Covers(j.boundary, ST_SetSRID(ST_MakePoint($1, $2), 4326));
```

✓ New routing uses latest gazetted boundaries without corrupting old records.

GOVERNANCE INVARIANT 01

Foreign Key Schema Integrity

The database enforces complaints.jurisdiction_version_id REFERENCES jurisdiction_versions(id) ON DELETE RESTRICT. Historical records can never be orphaned or modified by boundary changes.

✓ Cryptographic schema-level audit lock

GOVERNANCE INVARIANT 02

No In-Place Overwrites

Redistricting never executes an UPDATE on boundary geometry. A new version record is inserted with updated MultiPolygons, while the old version is flagged is_active = false.

✓ Zero historical data mutation

GOVERNANCE INVARIANT 03

Backward Re-Evaluation Safety

Historical complaints are never re-evaluated against new boundaries. Court and RTI inquiries retrieve the exact authority that held municipal jurisdiction at the moment of incident occurrence.

✓ Legal proof across municipal redistricting

KEY PRINCIPLE: "No in-place overwrite of historical routing context."

foreign_key: complaints.jurisdiction_version_id → immutable audit trail

Built. Deployed. Verified: Real Application Proof

Live application evidence: Sayyaji Rao Road pothole complaint executed on deterministic PostGIS logic.

ACTUAL COMPLAINT PHOTOGRAPH

VERIFIED REPOSITORY ASSET



Asset: Sayyaji Rao Road Asphalt Failure
Capture UI: Citizen Mobile PWA Interface
Sha256 Hash: 75d0c6cb42b882d0...

VERIFIED EXECUTION TRACE

AUTOMATED AUDIT RUN

- 01 POST /api/complaints

Validated intake payload, returns HTTP 201 Created

HTTP 201
- 02 Spatial Point Gen

ST_SetSRID(ST_MakePoint, 4326) creates standard geometry

EPSG:4326
- 03 ST_Covers Match

Contains in MCC_ZONE_1 MultiPolygon (Zone 1)

MATCHED
- 04 Department Queue

Automated SQL join assigns to MCC_ROADS with 48h SLA

MCC_ROADS
- 05 Socket.IO Broadcast

Real-time push to operator active queue and SLA tracker

PUSHED
- 06 Audit Committed

ROUTING_EXECUTED row sealed with MYS_2026_V2 stamp

SEALED

INCIDENT PROPERTIES & RESOLUTION RECORD

COMPLAINT ID:

HM-CIV-2026-000656

LOCATION:

Sayyaji Rao Rd / MCC Central Zone

CATEGORY:

POTHOLE (AI_SUGGESTED)

ACTIVE CADASTRE:

MYS_2026_V2 (ACTIVE)

RESPONSIBLE DEPARTMENT

MCC Roads & Infrastructure (MCC_ROADS)
Zero human triage required • Instant SLA dispatch

LIVE MVP DEPLOYED & OPERATIONAL

https://adaptive-civic-routing.vercel.app

Verified against actual Mysore City Corporation (MCC) & MUDA boundary demarcations.

AUTOMATED REGRESSION TEST

497/497 Checks Passing

End-to-end regression tests in testStage8.js and testStage14.js verify point creation, spatial containment, JWT authorization, Socket.IO broadcast, and SLA countdowns.

✓ 497/497 automated verification checks passing

AUDITABILITY & SLA TRACKING

Immutable Audit Ledger

Complaint HM-CIV-2026-000656 committed to immutable append-only ledger with 48h active SLA timer designed for accountable civic service workflows.

✓ Sealed audit row committed with V2 cadastre stamp

✓ PROVEN PRODUCTION RECORD: End-to-end civic complaint lifecycle verified live on deployed cloud infrastructure.

LIVE MVP DEPLOYED

What is Implemented — And What Comes Next

A transparent overview of current capabilities, real-world engineering constraints, and future scaling roadmap.

01 / SPATIAL COMPUTE

PostGIS Containment Engine

IMPLEMENTED MVP (WORKING NOW)

- PostGIS GiST R-Tree indexing for efficient point containment.
- Single-query binary ST_Covers check on active cadastre.
- PostgreSQL parameterized queries prevent SQL injection.
- Verified against real MCC and MUDA MultiPolygon boundaries.

```
CREATE INDEX ON jurisdictions USING GIST(boundary);
```

CURRENT REAL-WORLD LIMITATION

- Single managed PostgreSQL instance executes both writes and spatial checks.
- Complex municipal polygons with thousands of vertices increase CPU load under high concurrency.
- Browser map transfers full GeoJSON vectors, increasing payload size.

FUTURE SCALING ROADMAP

- Pre-computed Mapbox Vector Tiles (MVT) for ultra-fast browser rendering.
- Douglas-Peucker polygon simplification for edge mobile devices.
- Distributed spatial grid partitioning (Geohash / Quadtree) across Karnataka.

Metric: 98% candidate zone pruning via GiST R-Tree bounding-box index.

02 / FIELD CONNECTIVITY

Field Sync & Offline State

IMPLEMENTED MVP (WORKING NOW)

- Socket.IO WebSocket engine with exponential backoff reconnect.
- Graceful HTTP REST fallback if WebSocket connection drops.
- Server-side idempotency safeguards prevent duplicate ticket creation.
- Department rooms (dept:MCC_ROADS) isolate traffic.

```
io.connect({ reconnectionAttempts: 10, backoff: 'exp' });
```

CURRENT REAL-WORLD LIMITATION

- Field workers in urban basements or remote corridors lose cellular reception.
- Offline draft state currently retained only in browser session memory.
- High-resolution camera photos take several seconds on 2G/3G edge connections.

FUTURE SCALING ROADMAP

- Offline-first PWA with ServiceWorker background synchronization.
- Local IndexedDB storage queue for offline work orders and closure notes.
- On-device cryptographic timestamping before offline storage.

Target: Zero data loss on degraded municipal network links during field dispatch.

03 / DATABASE SCALE

Data Persistence & Auditing

IMPLEMENTED MVP (WORKING NOW)

- Normalized PostgreSQL schema with foreign keys (ON DELETE RESTRICT).
- Connection pooling via pg.Pool (max 20, idle timeout safeguards).
- Dedicated temporal cadastre tables (jurisdiction_versions).
- Append-only audit_logs ledger recording every state transition.

```
pool = new Pool({ max: 20, idleTimeoutMillis: 30000 });
```

CURRENT REAL-WORLD LIMITATION

- Single managed PostgreSQL instance handles writes and spatial queries.
- Audit log table grows linearly; unpartitioned table will slow queries over years.
- Analytical reporting queries compete with active complaint transactions.

FUTURE SCALING ROADMAP

- Read replicas dedicated to analytical dashboards and public ward reports.
- Declarative table partitioning by municipal zone and fiscal year.
- Change Data Capture (CDC) pipeline streaming audit events into cold storage.

Integrity: Transactional safety under monsoon peak complaint influx.

ENGINEERING TRANSPARENCY: We clearly distinguish working MVP capabilities from architectural roadmap items.

GREEN = PROVEN MVP

AMBER = CURRENT CONSTRAINT

BLUE = FUTURE SCALING

From Ward-Level Pilot to City-Wide Civic Intelligence

A clear roadmap for municipal scale, coupled with ethical AI governance and complete project disclosure.

PHASED MUNICIPAL SCALING ROADMAP

PHASE 1 — CURRENT MVP

DEPLOYED & LIVE

- Ward/zone PostGIS routing foundation (MCC Zone 1 vs MUDA).
- Live Vercel frontend + Render backend & PostGIS deployment.
- Realtime operator queue with Socket.IO & 48h SLA countdowns.
- Temporal auditability with versioned cadastre (MYS_2026_V1/V2).
- Multi-role RBAC security for Citizens, Operators, and Admins.

✓ Verified live production system

PHASE 2 — MUNICIPAL GIS INTEGRATION

NEXT 6 MONTHS

- Ingest complete MCC 65-ward cadastre & MUDA layout boundaries.
- Multi-agency cross-routing (MCC, MUDA, CESC, Water Supply).
- Field crew mobile app with offline-first work orders & photo sync.
- Administrative boundary redistricting management console.

✓ City-wide enterprise expansion

PHASE 3 — CITIZEN ACCESSIBILITY

NEXT 12 MONTHS

- Kannada vernacular voice/text intake for non-literate citizens.
- WhatsApp Business API & automated telephone IVR reporting.
- Automated duplicate complaint clustering & hotspot detection.
- Public municipal performance scorecards & ward health rankings.

✓ Universal civic inclusion

Vision: A single civic routing standard for all Karnataka municipal bodies.

RESPONSIBLE AI DISCLOSURE & GOVERNANCE

Google Gemini 1.5 API (Assistive AI):

Natural language categorization & structured entity suggestion ONLY. Zero jurisdiction, dispatch, or administrative decision authority.

Antigravity AI (Development & Testing):

Full-stack development acceleration, automated schema validation, and comprehensive test suite execution across 497 automated verification checks.

Designed for Accountable Civic Workflows:

Immutable SLA timestamps adhere to transparent civic service standards for municipal accountability.

✓ Core Law: Coordinate → Polygon → Jurisdiction → Department remains deterministic GIS jurisdiction resolution PostGIS logic.

SUBMISSION DETAILS

Team Spark • HM26-E9C5

HackMysuru 1.0 — Problem Track: Routing

Team Members: Hemanth N • Karibasappa KL • Pooja M • Soumya Doddamani

Live MVP: adaptive-civic-routing.vercel.app

GitHub: github.com/kkaribasappakl-eng/adaptive-civic-routing

Decision Log: Verified Hackathon Architectural Decision Record (decision-log.pdf)

TEAM SPARK | HM26-E9C5

"Report once. Route correctly. Resolve with accountability."

Adaptive Civic Routing Intelligence System (ACRIS) • HackMysuru 1.0 Final Submission